



Hochschule Aalen
UNIVERSITY OF APPLIED SCIENCES

Machine Learning & Deep Learning

00: COURSE INFORMATION



Lecturer

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Fields of interest:

- Maschine Learning / Deep Learning
- Visual Analytics
- Medical Artificial Intelligence
- Explainable AI

Research:

- Lung Cancer Classification & Detection
- Hirachical Classification
- Sleep apnea, delirium, etc.

If you are interested in this research area or in applying ML/DL to any domain, feel free to ask about potential Master's thesis topics, or propose your own idea.

Course Information

- Elements of the course:
 - 14 lecture dates
 - Approx. 5 Lab-Sessions
 - 60 hours lecture, 90 hours self study
- Course Materials
 - Will be provided via canvas
 - Lecture notes in slide format
 - May be supplemented by lecture notes, if you are not present obtain the notes from other students
 - Additional documents for self-study (Python Introduction, Machine Learning with Python, etc.)

Exam

- Grade
 - Written exam (90 min)
 - Theoretical part: Everything we learn in the lectures
 - Practical part: Interpretation and coding of python programming code (yes, writing code on paper)
 - Weighting: tbd
- Overarching learning objectives:
 - Understanding the basics of Machine Learning and Deep Learning, in particular, predictive models and their special properties.
 - Ability to apply machine learning to own problems (e.g. in the course of Machine Learning Research)
 - Ability to communicate results professionally, including clear reporting of assumptions, limitations, and reproducibility (e.g., documenting data, settings, and model choices).

General course structure

- Intro + Workflow
- Classification + Regression (Tabular)
- Clustering
- Deep Learning Foundations
- DL for Vision (CNNs)
- DL for Text (Transformers)
- Self-Supervised Pretraining + Fine-tuning
- Generative AI / GenAI (LLMs + diffusion)
- Time Series (classical + DL model)
- Semi-Supervised + Reinforcement Learning overview